

Digonta Das

AI Engineer | Computer Vision Engineer | Web Developer

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TECHNICAL SKILLS

ML & Deep Learning: PyTorch, TensorFlow, CNN, 3D CNN, ResNet, U-Net, MobileNetV2, Transfer Learning, XGBoost, LoRA, QLoRA, ONNX

Computer Vision: Segmentation, Object Detection, Volumetric Analysis, HU Normalization, Data Augmentation

NLP & LLMs: Hugging Face, Sentence Transformers, Prompt Engineering, RAG, LangChain, LLM APIs (OpenAI, Gemini, Groq, Whisper)

Backend & Infra: FastAPI, Node.js, Express.js, Docker, GitHub Actions, Vercel, Render, Hugging Face Spaces

Databases & Vector Stores: PostgreSQL, Supabase, ChromaDB, pgvector, MongoDB, SQLite

Frontend & Mobile: React, React Native (Expo), Next.js, Tailwind CSS | **Languages:** Python, JavaScript, C, Kotlin

PROFESSIONAL EXPERIENCE

AI Developer Intern March 2026 – June 2026

IntelliVerseRemote

- Cut transcription pipeline response time by 40% by redesigning request batching for a Whisper API speech-to-text system; built a FastAPI inference backend reducing API failure rate to under 2% and shipped 3 LLM integrations into client products within the first month.

RESEARCH

Efficient 3D Tiled CNN for Coronary Artery Stenosis Detection Ongoing

Undergraduate Thesis (Team Lead) | BRAC University

- End-to-end 3D CNN segmentation pipeline on 121 MRI + 160 CT scans (3.5 GB/scan) self-collected from Ibrahim Cardiac & Bangladesh Medical hospitals, benchmarked against ImageCAS.
- Designed volumetric tiling for memory-efficient CT processing; engineered HU clipping/normalization and Hybrid Combo Loss (Dice + BCE) to resolve empty-mask collapse; outputs structured cardiac risk scores for clinicians.

Efficient SE-Mobile UNet for Cross-Vendor Cardiac MRI Segmentation

github.com/DigontaDas/Hybrid-Model-on-Efficient-SE-Net

- Hybrid MobileNetV2 + SE-attention CNN at 2.2M parameters; zero-shot cross-vendor eval: DSC 0.8712 on ACDC, DSC 0.7512 on unseen M&Ms dataset (generalization gap: 0.12).

KEY PROJECTS

MaSheba AI – Offline Maternal Health Platform

React Native, FastAPI, ONNX, XGBoost, Supabase | [GitHub](#) | [Docs](#)

- Team Lead of 5-person team; Infinity AI BuildFest 2026 Finalist — offline-first app targeting 3M+ expecting mothers and 60K+ CHWs in rural Bangladesh; XGBoost risk model exported to ONNX (1 MB) achieving <200ms on-device inference on Android 8 with zero internet dependency.
- Outbox-first SQLite sync stress-tested at 50 concurrent events at 100% deduplication; cascading Bangla AI chat (Groq → Gemini → offline) with 6-stage safety filter; Next.js admin dashboard backed by Supabase + pgvector.

Clarity – B2B Supply Chain Finance Platform

PostgreSQL, Express.js, React, Node.js, Backend Routing, API Integration, ERP Systems, Financial Algorithms | [GitHub](#)

- Engineered a B2B supply chain finance platform with an invoice marketplace, implementing Express.js backend routes with locking mechanisms to guarantee secure, non-duplicated transaction processing.
- Built a real-time discount rate calculator and integrated ERP system data structures to streamline financial operations for enterprise clients; architected a secure KYB (Know Your Business) verification vault for business authentication and compliance data.
- Implemented invoice confirmation and digital acknowledgment workflows across frontend and backend APIs; managed backend infrastructure on Supabase powering real-time dashboards with seamless data synchronization.

Movie Recommendation AI – RAG Pipeline

ChromaDB, Sentence Transformers, LLaMA, FastAPI, React | [GitHub](#)

- Full RAG pipeline over 3,000+ movies using Sentence Transformers + ChromaDB; LLM re-ranking via Ollama (LLaMA) with modular FastAPI backend and React/Vite frontend, fully local at zero API cost.

EDUCATION

BRAC University Dhaka, Bangladesh

B.Sc. in Computer Science and Engineering | GPA: 3.6 / 4.0 Expected Jan 2027

- Coursework: Artificial Intelligence, Computer Vision, Image Processing, Computer Graphics, Linear Algebra, Operating Systems, Computer Architecture

AWARDS & RECOGNITION

- **Edupro Global Green Talent Awards 2026** – Round 1 Qualifier & Round 2 Participant May–Jun 2026
Team Jagotic Moho | In association with the University of Leeds, UK – demonstrating creativity, innovation, and commitment to sustainability.
- **Infinity AI BuildFest 2026** – Finalist, Team DareDevil | Offline-first maternal health AI platform (MaSheba) for rural Bangladesh 2026